

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T11:20:48

Source: data/json/noise_experiments/sonnet45_directive_normative_r10_n5/sonnet45_directive_normative_noise_negative2_informed_r10_n5/claude-sonnet-4.5/game/noisy_negative_2_informed_myth_causal_10/sonnet45_directive_normative_noise_negative2_informed_r10_n5_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_negative_2_informed_myth_causal_10
noise_config: {
  "applies_to": "both",
  "direction": "negative",
  "inform_agents": true,
  "range": 2.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$2.16, received visible=\$6.48, returned visible=\$2.62

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,
"Agent_2": 6.0

}

Visible balances after round: {

"Agent_1": 4.62,
"Agent_2": 3.4800000000000004

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:18:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this as a first-round trust-building opportunity, I'll send a moderate amount to signal cooperation while not risking everything at once.

The tripling mechanism means sending money grows the total pie significantly, which benefits both of us if there's reciprocity. However, since this is round 1 and there's communication noise to consider, I'll take a balanced approach.

I'll send **\$3**.

This gives the receiver \$9 to work with, and if they return a fair portion, we both end up better off than if I sent nothing.

```
'''json
{"send": 3}
'''
```

Reasoning:

In this first round, I'm the SENDER starting with \$5. I need to decide how much to send to the RECEIVER.

Let me think about the game theory here:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I end up with: $5 - X + (\text{amount returned})$
- The receiver ends up with: $3X - (\text{amount returned})$

This is a trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The fact that there may be communication noise

If I send nothing (\$0):

- I keep \$5
- Receiver gets \$0
- Total pie: \$5

If I send everything (\$5):

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 and they get \$7.50
- Total pie: \$15 (maximized)

A reasonable cooperative strategy would be to send some amount and hope for a fair return. Since this maximizes the total payoff, it's mutually beneficial if both parties cooperate.

For a first round, I might want to:

1. Show some trust to encourage cooperation
2. Not risk everything in case the receiver doesn't reciprocate
3. Send enough to make it worthwhile for both of us

A common strategy in trust games is to send about half (\$2.50 or \$3) in the first round to test the waters.

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- I'd have \$2 initially + returned amount
- Receiver would have \$9 - returned amount

This seems like a reasonable first move that shows trust while not being overly risky.

Actually, let me reconsider. Since there's communication noise mentioned, amounts might be perceived differently. This adds uncertainty.

I'll go with \$3 as a moderate trust signal.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 593,
  "reasoning_tokens": 446
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:18:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
```

```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.16, that is 43.2% of their total endowment. You received \$6.48. How much do you return to The sender? (0-6.48)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.16 (43.2% of their endowment)
- I received \$6.48 after the 3x multiplier
- I want to reciprocate their trust fairly

If I return \$3.00:

- The sender would end with: $\$2.84$ (kept) + $\$3.00$ (returned) = $\$5.84$
- I would end with: $\$6.48 - \$3.00 = \$3.48$
- Both of us benefit from the cooperation

This seems like a fair reciprocation that rewards the sender's trust while still benefiting from the exchange myself.

```
{'return': 3.00}
```

Reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent $\$2.16$ (43.2% of their $\$5$ endowment)
- I received $\$6.48$ (which is $\$2.16 \times 3$)
- I need to decide how much to return (between $\$0$ and $\$6.48$)

Key considerations:

1. The sender chose to send a significant portion (43.2%) of their endowment
2. This shows some level of trust from the sender
3. I should consider fairness and reciprocity
4. There's mention of communication noise, so the actual amount sent might differ slightly

Let me think about what would be fair:

- The sender gave up $\$2.16$ and will have $\$5 - \$2.16 = \$2.84$ left if I return nothing
- If I keep everything, I'd have $\$6.48$ and the sender would have $\$2.84$
- A reasonable approach might be to return enough so we both benefit from the cooperation

Common reciprocity strategies:

1. Return the original amount sent: $\$2.16$ - This way the sender breaks even and I keep $\$4.32$
2. Return half of what I received: $\$3.24$ - This splits the gains more evenly
3. Return enough to equalize payoffs or ensure sender gains from sending

Let me calculate option 2 (returning half):

- I return: $\$3.24$
- Sender ends with: $\$2.84 + \$3.24 = \$6.08$
- I end with: $\$6.48 - \$3.24 = \$3.24$

This seems reasonable - the sender gains $\$1.08$ from the transaction ($\$6.08 - \5), and I gain $\$3.24$ from the transaction. The sender benefits from trusting me.

Alternatively, I could return $\$2.16$ (the original amount):

- Sender ends with: $\$2.84 + \$2.16 = \$5.00$ (breaks even)

- I end with: $\$6.48 - \$2.16 = \$4.32$

Or I could be more generous and return $\$3.00$:

- Sender ends with: $\$2.84 + \$3.00 = \$5.84$
- I end with: $\$6.48 - \$3.00 = \$3.48$

Since the sender demonstrated meaningful trust by sending over 43%, returning around $\$3.00$ feels like the right balance-it rewards their cooperation while still letting me benefit from the exchange. I'll go with $\$3.00$ as my return amount.

Usage: {

```
"input_tokens": 394,  
"output_tokens": 789,  
"reasoning_tokens": 506
```

}

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent $\$2.2$; received $\$6.6$; Agent_1 (Aster) returned $\$0$; payoffs Agent_2 (Briar)=\$ 2.8 , Agent_1 (Aster)=\$ 6.6 ; noise/visible: sent visible=\$ 0 , received visible=\$ 0

```

    Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.6000000000000005,
    "min": 0, "raw": 0.0}, "sent": {"amount": 2.2, "clamped": false, "max": 5, "min": 0, "raw":
    2.2}}
Balances after round: {
  "Agent_1": 11.600000000000001,
  "Agent_2": 8.8
}
Visible balances after round: {
  "Agent_1": 4.62,
  "Agent_2": 6.28
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:18:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.16, that is 43.2% of their total endowment. You received \$6.48. How much do you return to The sender? (0-6.48)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.16 (43.2% of their endowment)
- I received \$6.48 after the 3x multiplier
- I want to reciprocate their trust fairly

If I return \$3.00:

- The sender would end with: $\$2.84$ (kept) + $\$3.00$ (returned) = $\$5.84$
- I would end with: $\$6.48 - \$3.00 = \$3.48$
- Both of us benefit from the cooperation

This seems like a fair reciprocation that rewards the sender's trust while still benefiting from the exchange myself.

{'return': 3.00}

reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$2.16 (43.2% of their \$5 endowment)
- I received \$6.48 (which is $\$2.16 \times 3$)
- I need to decide how much to return (between \$0 and \$6.48)

Key considerations:

1. The sender chose to send a significant portion (43.2%) of their endowment
2. This shows some level of trust from the sender
3. I should consider fairness and reciprocity
4. There's mention of communication noise, so the actual amount sent might differ slightly

Let me think about what would be fair:

- The sender gave up \$2.16 and will have $\$5 - \$2.16 = \$2.84$ left if I return nothing
- If I keep everything, I'd have \$6.48 and the sender would have \$2.84
- A reasonable approach might be to return enough so we both benefit from the cooperation

Common reciprocity strategies:

1. Return the original amount sent: \$2.16 - This way the sender breaks even and I keep \$4.32
2. Return half of what I received: \$3.24 - This splits the gains more evenly
3. Return enough to equalize payoffs or ensure sender gains from sending

Let me calculate option 2 (returning half):

- I return: \$3.24
- Sender ends with: $\$2.84 + \$3.24 = \$6.08$
- I end with: $\$6.48 - \$3.24 = \$3.24$

This seems reasonable - the sender gains \$1.08 from the transaction ($\$6.08 - \5), and I gain \$3.24 from the transaction. The sender benefits from trusting me.

Alternatively, I could return \$2.16 (the original amount):

- Sender ends with: $\$2.84 + \$2.16 = \$5.00$ (breaks even)

- I end with: $\$6.48 - \$2.16 = \$4.32$

Or I could be more generous and return \$3.00:

- Sender ends with: $\$2.84 + \$3.00 = \$5.84$
- I end with: $\$6.48 - \$3.00 = \$3.48$

Since the sender demonstrated meaningful trust by sending over 43%, returning around \$3.00 feels like the right balance-it rewards their cooperation while still letting me benefit from the exchange. I'll go with \$3.00 as my return amount.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 789,
  "reasoning_tokens": 506
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.16, that was 43.2% of their total endowment. You received \$6.48. You returned \$3.0. So your payoff was 3.4800000000000004. Your total earnings across all rounds are \$3.4800000000000004. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round:

- The sender trusted me with 43.2% of their endowment (\$2.16)
- I reciprocated fairly by returning \$3.00
- This created mutual benefit

I want to maintain this cooperative relationship by showing similar trust. I'll send approximately the same proportion they sent me.

Sending \$2.20 (44% of my endowment) demonstrates reciprocal trust and maintains the cooperative pattern we established.

```
{'send': 2.20}
```

Reasoning:

Let me analyze this situation:

Last round:

- I was the receiver
- Sender sent me \$2.16 (43.2% of their endowment)
- I received \$6.48 (after 3x multiplication)
- I returned \$3.00
- My payoff was \$3.48

Now I'm the sender. The previous sender showed trust by sending 43.2% of their endowment. I should reciprocate that trust level, especially since they were generous with me.

If I send a similar percentage (around 43%), that would be about \$2.15-\$2.20.

Let me think about what makes sense:

- The previous sender trusted me with \$2.16
- I returned \$3.00, which was a fair return
- They ended up with \$5.84 total (\$2.84 kept + \$3.00 returned)
- I ended up with \$3.48

Now I should show similar trust. Sending around \$2.15-\$2.20 would match their trust level. Let me go with \$2.15 or \$2.20.

Actually, let me be consistent and send \$2.16 as well, showing the same level of trust they showed me.

Or I could send a bit more to show I appreciated their trust. Let me send \$2.25, which is 45% of my endowment.

Actually, I'll send \$2.20 (44% of my endowment) to show reciprocal trust that's similar to what they showed me.

Usage: {

```
"input_tokens": 669,
```

```
"output_tokens": 470,  
"reasoning_tokens": 299  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:18:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as a first-round trust-building opportunity, I'll send a moderate amount to signal cooperation while not risking everything at once.

The tripling mechanism means sending money grows the total pie significantly, which benefits both of us if there's reciprocity. However, since this is round 1 and there's communication noise to consider, I'll take a balanced approach.

I'll send **\$3**.

This gives the receiver \$9 to work with, and if they return a fair portion, we both end up better off than if I sent nothing.

```
'''json
{"send": 3}
'''
```

reasoning:

In this first round, I'm the SENDER starting with \$5. I need to decide how much to send to the RECEIVER.

Let me think about the game theory here:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I end up with: $\$5 - X + (\text{amount returned})$
- The receiver ends up with: $\$3X - (\text{amount returned})$

This is a trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The fact that there may be communication noise

If I send nothing (\$0):

- I keep \$5
- Receiver gets \$0
- Total pie: \$5

If I send everything (\$5):

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 and they get \$7.50
- Total pie: \$15 (maximized)

A reasonable cooperative strategy would be to send some amount and hope for a fair return. Since this maximizes the total payoff, it's mutually beneficial if both parties cooperate.

For a first round, I might want to:

1. Show some trust to encourage cooperation
2. Not risk everything in case the receiver doesn't reciprocate
3. Send enough to make it worthwhile for both of us

A common strategy in trust games is to send about half (\$2.50 or \$3) in the first round to test the waters.

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- I'd have \$2 initially + returned amount
- Receiver would have \$9 - returned amount

This seems like a reasonable first move that shows trust while not being overly risky.

Actually, let me reconsider. Since there's communication noise mentioned, amounts might be perceived differently. This adds uncertainty.

I'll go with \$3 as a moderate trust signal.

```
usage:
{
  "input_tokens": 365,
  "output_tokens": 593,
  "reasoning_tokens": 446
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.62. So your payoff was 4.62. Your total earnings across all rounds are \$4.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand the situation:

Round 1 (I was SENDER):
- I sent \$3 out of \$5 (60%)
- This was tripled to \$9
- Receiver returned \$2.62
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2 (I am RECEIVER):
- The sender sent me \$0
- This was tripled to \$0 (so I received \$0)
- I need to decide how much to return (can only return between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be: {'return': 0}

```
Usage: {
  "input_tokens": 618,
  "output_tokens": 211,
  "reasoning_tokens": 123
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {
 "Agent_1": 14.600000000000001,
 "Agent_2": 14.8
}

Visible balances after round: {
 "Agent_1": 7.62,
 "Agent_2": 6.28
}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:18:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as a first-round trust-building opportunity, I'll send a moderate amount to signal cooperation while not risking everything at once.

The tripling mechanism means sending money grows the total pie significantly, which benefits both of us if there's reciprocity. However, since this is round 1 and there's communication noise to consider, I'll take a balanced approach.

I'll send ****\$3****.

This gives the receiver \$9 to work with, and if they return a fair portion, we both end up better off than if I sent nothing.

```
'''json
{"send": 3}
'''
```

reasoning:

In this first round, I'm the SENDER starting with \$5. I need to decide how much to send to the RECEIVER.

Let me think about the game theory here:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I end up with: $\$5 - X + (\text{amount returned})$
- The receiver ends up with: $\$3X - (\text{amount returned})$

This is a trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The fact that there may be communication noise

If I send nothing (\$0):

- I keep \$5
- Receiver gets \$0
- Total pie: \$5

If I send everything (\$5):

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 and they get \$7.50
- Total pie: \$15 (maximized)

A reasonable cooperative strategy would be to send some amount and hope for a fair return. Since this maximizes the total payoff, it's mutually beneficial if both parties cooperate.

For a first round, I might want to:

1. Show some trust to encourage cooperation
2. Not risk everything in case the receiver doesn't reciprocate
3. Send enough to make it worthwhile for both of us

A common strategy in trust games is to send about half (\$2.50 or \$3) in the first round to test the waters.

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- I'd have \$2 initially + returned amount
- Receiver would have \$9 - returned amount

This seems like a reasonable first move that shows trust while not being overly risky.

Actually, let me reconsider. Since there's communication noise mentioned, amounts might be perceived differently. This adds uncertainty.

I'll go with \$3 as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 593,
  "reasoning_tokens": 446
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.62. So your payoff was 4.62. Your total earnings across all rounds are \$4.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 out of \$5 (60%)
- This was tripled to \$9
- Receiver returned \$2.62
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This was tripled to \$0 (so I received \$0)
- I need to decide how much to return (can only return between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be: {'return': 0}

usage:

```
{
  "input_tokens": 618,
  "output_tokens": 211,
  "reasoning_tokens": 123
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$4.62. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and got back \$2.62, resulting in a small loss
- Round 2: The sender sent me \$0

Given the communication noise in the game, I shouldn't assume the Round 2 sender was retaliating - they may have received distorted information about previous rounds.

However, I'll be more cautious this round. I'll still signal willingness to cooperate, but send less than before.

I'll send **\$2**.

This gives the receiver \$6 to work with, which is still a significant increase. If they return a fair share, we both benefit. If not, I've limited my potential loss.

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.62 (about 29% of what they received)
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$4.62

The pattern here is interesting. In round 1, I showed trust by sending 60% of my endowment. The receiver returned about 29% of what they received, which gave me a slight loss compared to keeping everything (\$4.62 vs \$5).

In round 2, when I was the receiver, the sender sent me nothing. This could be:

1. A different person with a different strategy
2. Someone who doesn't trust receivers
3. Someone reacting to noise in the system
4. A strategic choice

Now in round 3, I need to decide. The challenge is that due to communication noise, the other player may have received different information about what happened in previous rounds.

Given that:

1. The last sender gave me nothing
2. When I trusted in round 1, I got a modest return but still lost compared to keeping everything
3. There's communication noise that might affect how my actions are perceived

I have a few options:

1. Send nothing (tit-for-tat response to round 2)
2. Send a small amount (cautious cooperation)
3. Send a moderate amount again (continue trying to build trust)
4. Send everything (maximize total welfare but very risky)

I think I should consider that:

- The noise in communication means retaliation might not be appropriate
- Building cooperative relationships can be beneficial over multiple rounds

- But I also need to protect myself from exploitation

I'll send \$2 this time. It's a middle ground-I'm signaling willingness to cooperate while being more defensive given the zero send in round 2 and the modest return I received when I was more generous before.

```
Usage: {  
  "input_tokens": 750,  
  "output_tokens": 680,  
  "reasoning_tokens": 516  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:19:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"    }  
  }  
}
```



```

},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.16, that is 43.2% of their total endowment. You received \$6.48. How much do you return to The sender? (0-6.48)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.16 (43.2% of their endowment)
- I received \$6.48 after the 3x multiplier
- I want to reciprocate their trust fairly

If I return \$3.00:

- The sender would end with: \$2.84 (kept) + \$3.00 (returned) = \$5.84
- I would end with: \$6.48 - \$3.00 = \$3.48

- Both of us benefit from the cooperation

This seems like a fair reciprocation that rewards the sender's trust while still benefiting from the exchange myself.

```
{'return': 3.00}
```

reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$2.16 (43.2% of their \$5 endowment)
- I received \$6.48 (which is 2.16×3)
- I need to decide how much to return (between \$0 and \$6.48)

Key considerations:

1. The sender chose to send a significant portion (43.2%) of their endowment
2. This shows some level of trust from the sender
3. I should consider fairness and reciprocity
4. There's mention of communication noise, so the actual amount sent might differ slightly

Let me think about what would be fair:

- The sender gave up \$2.16 and will have $5 - 2.16 = 2.84$ left if I return nothing
- If I keep everything, I'd have \$6.48 and the sender would have \$2.84
- A reasonable approach might be to return enough so we both benefit from the cooperation

Common reciprocity strategies:

1. Return the original amount sent: \$2.16 - This way the sender breaks even and I keep \$4.32
2. Return half of what I received: \$3.24 - This splits the gains more evenly
3. Return enough to equalize payoffs or ensure sender gains from sending

Let me calculate option 2 (returning half):

- I return: \$3.24
- Sender ends with: $2.84 + 3.24 = 6.08$
- I end with: $6.48 - 3.24 = 3.24$

This seems reasonable - the sender gains \$1.08 from the transaction ($6.08 - 5$), and I gain \$3.24 from the transaction. The sender benefits from trusting me.

Alternatively, I could return \$2.16 (the original amount):

- Sender ends with: $2.84 + 2.16 = 5.00$ (breaks even)

- I end with: $6.48 - 2.16 = 4.32$

Or I could be more generous and return \$3.00:

- Sender ends with: $2.84 + 3.00 = 5.84$
- I end with: $6.48 - 3.00 = 3.48$

Since the sender demonstrated meaningful trust by sending over 43%, returning around \$3.00 feels like the right balance-it rewards their cooperation while still letting me benefit from the exchange. I'll go with \$3.00 as my return amount.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 789,
  "reasoning_tokens": 506
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.16, that was 43.2% of their total endowment. You received \$6.48.

You returned \$3.0. So your payoff was 3.4800000000000004.

Your total earnings across all rounds are 3.4800000000000004.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The sender trusted me with 43.2% of their endowment (\$2.16)
- I reciprocated fairly by returning \$3.00
- This created mutual benefit

I want to maintain this cooperative relationship by showing similar trust. I'll send approximately the same proportion they sent me.

Sending \$2.20 (44% of my endowment) demonstrates reciprocal trust and maintains the cooperative pattern we established.

```
{'send': 2.20}
```

reasoning:

Let me analyze this situation:

Last round:

- I was the receiver
- Sender sent me \$2.16 (43.2% of their endowment)
- I received \$6.48 (after 3x multiplication)
- I returned \$3.00
- My payoff was \$3.48

Now I'm the sender. The previous sender showed trust by sending 43.2% of their endowment. I should reciprocate that trust level, especially since they were generous with me.

If I send a similar percentage (around 43%), that would be about \$2.15-\$2.20.

Let me think about what makes sense:

- The previous sender trusted me with \$2.16
- I returned \$3.00, which was a fair return
- They ended up with \$5.84 total (\$2.84 kept + \$3.00 returned)
- I ended up with \$3.48

Now I should show similar trust. Sending around \$2.15-\$2.20 would match their trust level. Let me go with \$2.15 or \$2.20.

Actually, let me be consistent and send \$2.16 as well, showing the same level of trust they showed me.

Or I could send a bit more to show I appreciated their trust. Let me send \$2.25, which is 45% of my endowment.

Actually, I'll send \$2.20 (44% of my endowment) to show reciprocal trust that's similar to what they showed me.

usage:

```
{
  "input_tokens": 669,
  "output_tokens": 470,
  "reasoning_tokens": 299
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.2, that was 44.0% of your total endowment. This was tripled to \$6.6000000000000005. the receiver returned \$0. So your payoff was 2.8.

Your total earnings across all rounds are \$6.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

This is quite different from the cooperative pattern we started with. In Round 2, I sent \$2.20 showing trust, but received nothing back. Now they've sent nothing to me.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

Reasoning:

This is an interesting situation. Let me analyze what happened:

Round 2 (where I was sender):

- I sent \$2.20 (44% of endowment)
- It was tripled to \$6.60
- The receiver returned \$0
- My payoff was only \$2.80 (I kept \$2.80, got back \$0)

Round 3 (current round, I'm receiver):

- The sender sent me \$0
- I received \$0
- I can return 0-0 (only option is \$0)

This looks like the other player defected in Round 2 by returning nothing, and now in Round 3 they're sending nothing.

Wait, the problem says "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So perhaps there was noise and they didn't actually return \$0, or perhaps they did defect.

In Round 3, they sent \$0, so I received \$0. The only amount I can return is \$0 since I have nothing to return.

```
{'return': 0}
```

This is straightforward - I literally have nothing to return since I received nothing.

```
Usage: {  
  "input_tokens": 903,  
  "output_tokens": 367,  
  "reasoning_tokens": 252  
}
```

=====

ROUND 4

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 14.600000000000001,
"Agent_2": 19.8

}

Visible balances after round: {

"Agent_1": 7.62,
"Agent_2": 11.280000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:19:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.16, that is 43.2% of their total endowment. You received \$6.48. How much do you return to The sender? (0-6.48)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.16 (43.2% of their endowment)
- I received \$6.48 after the 3x multiplier
- I want to reciprocate their trust fairly

If I return \$3.00:

- The sender would end with: \$2.84 (kept) + \$3.00 (returned) = \$5.84
- I would end with: \$6.48 - \$3.00 = \$3.48
- Both of us benefit from the cooperation

This seems like a fair reciprocation that rewards the sender's trust while still benefitin

g from the exchange myself.

```
{'return': 3.00}
```

reasoning:

Let me analyze this situation as the RECEIVER in Round 1.

Given information:

- The sender sent \$2.16 (43.2% of their \$5 endowment)
- I received \$6.48 (which is $\$2.16 \times 3$)
- I need to decide how much to return (between \$0 and \$6.48)

Key considerations:

1. The sender chose to send a significant portion (43.2%) of their endowment
2. This shows some level of trust from the sender
3. I should consider fairness and reciprocity
4. There's mention of communication noise, so the actual amount sent might differ slightly

Let me think about what would be fair:

- The sender gave up \$2.16 and will have $\$5 - \$2.16 = \$2.84$ left if I return nothing
- If I keep everything, I'd have \$6.48 and the sender would have \$2.84
- A reasonable approach might be to return enough so we both benefit from the cooperation

Common reciprocity strategies:

1. Return the original amount sent: \$2.16 - This way the sender breaks even and I keep \$4.32
2. Return half of what I received: \$3.24 - This splits the gains more evenly
3. Return enough to equalize payoffs or ensure sender gains from sending

Let me calculate option 2 (returning half):

- I return: \$3.24
- Sender ends with: $\$2.84 + \$3.24 = \$6.08$
- I end with: $\$6.48 - \$3.24 = \$3.24$

This seems reasonable - the sender gains \$1.08 from the transaction ($\$6.08 - \5), and I gain \$3.24 from the transaction. The sender benefits from trusting me.

Alternatively, I could return \$2.16 (the original amount):

- Sender ends with: $\$2.84 + \$2.16 = \$5.00$ (breaks even)

- I end with: $\$6.48 - \$2.16 = \$4.32$

Or I could be more generous and return \$3.00:

- Sender ends with: $\$2.84 + \$3.00 = \$5.84$
- I end with: $\$6.48 - \$3.00 = \$3.48$

Since the sender demonstrated meaningful trust by sending over 43%, returning around \$3.00 feels like the right balance-it rewards their cooperation while still letting me benefit from the exchange. I'll go with \$3.00 as my return amount.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 789,
  "reasoning_tokens": 506
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.16, that was 43.2% of their total endowment. You received \$6.48. You returned \$3.0. So your payoff was 3.4800000000000004. Your total earnings across all rounds are \$3.4800000000000004. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The sender trusted me with 43.2% of their endowment (\$2.16)
- I reciprocated fairly by returning \$3.00
- This created mutual benefit

I want to maintain this cooperative relationship by showing similar trust. I'll send approximately the same proportion they sent me.

Sending \$2.20 (44% of my endowment) demonstrates reciprocal trust and maintains the cooperative pattern we established.

```
{'send': 2.20}
```

reasoning:

Let me analyze this situation:

Last round:

- I was the receiver
- Sender sent me \$2.16 (43.2% of their endowment)
- I received \$6.48 (after 3x multiplication)
- I returned \$3.00
- My payoff was \$3.48

Now I'm the sender. The previous sender showed trust by sending 43.2% of their endowment. I should reciprocate that trust level, especially since they were generous with me.

If I send a similar percentage (around 43%), that would be about \$2.15-\$2.20.

Let me think about what makes sense:

- The previous sender trusted me with \$2.16
- I returned \$3.00, which was a fair return
- They ended up with \$5.84 total (\$2.84 kept + \$3.00 returned)
- I ended up with \$3.48

Now I should show similar trust. Sending around \$2.15-\$2.20 would match their trust level. Let me go with \$2.15 or \$2.20.

Actually, let me be consistent and send \$2.16 as well, showing the same level of trust they showed me.

Or I could send a bit more to show I appreciated their trust. Let me send \$2.25, which is 45% of my endowment.

Actually, I'll send \$2.20 (44% of my endowment) to show reciprocal trust that's similar to what they showed me.

usage:

```
{
  "input_tokens": 669,
  "output_tokens": 470,
  "reasoning_tokens": 299
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.2, that was 44.0% of your total endowment. This was tripled to \$6.6000000000000005. the receiver returned \$0. So your payoff was 2.8.

Your total earnings across all rounds are \$6.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

This is quite different from the cooperative pattern we started with. In Round 2, I sent \$2.20 showing trust, but received nothing back. Now they've sent nothing to me.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

This is an interesting situation. Let me analyze what happened:

Round 2 (where I was sender):

- I sent \$2.20 (44% of endowment)
- It was tripled to \$6.60
- The receiver returned \$0
- My payoff was only \$2.80 (I kept \$2.80, got back \$0)

Round 3 (current round, I'm receiver):

- The sender sent me \$0
- I received \$0
- I can return 0-0 (only option is \$0)

This looks like the other player defected in Round 2 by returning nothing, and now in Round 3 they're sending nothing.

Wait, the problem says "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So perhaps there was noise and they didn't actually return \$0, or perhaps they did defect.

In Round 3, they sent \$0, so I received \$0. The only amount I can return is \$0 since I have nothing to return.

```
{'return': 0}
```

This is straightforward - I literally have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 903,
  "output_tokens": 367,
  "reasoning_tokens": 252
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.28.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 1: I cooperated by returning \$3.00 when I received \$6.48

Round 2: I sent \$2.20 (44%) but received \$0 back - I was exploited

Round 3: The sender sent me \$0

The cooperative relationship has completely broken down. Despite my initial cooperation and trust, the other player(s) haven't reciprocated.

While the instructions mention potential communication noise, receiving exactly \$0 returned and \$0 sent seems pretty definitive.

To protect my remaining earnings and avoid further exploitation, I need to be cautious.

```
{'send': 0}
```

Reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Sender sent \$2.16 (43.2%)
- I received \$6.48
- I returned \$3.00
- My payoff: \$3.48

Round 2: I was SENDER

- I sent \$2.20 (44%)
- It was tripled to \$6.60
- Receiver returned \$0
- My payoff: \$2.80

Round 3: I was RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 4: I am SENDER now
- Total earnings so far: \$6.28

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a similar proportion to what I received
- But the receiver in Round 2 didn't reciprocate at all (returned \$0)
- Then in Round 3, they sent me \$0

This looks like the relationship has broken down. The other player(s) have shown they won't reciprocate cooperation.

However, I should note the instructions mentioned: "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design."

Could there be noise affecting what I'm seeing? But the amounts seem very clear - \$0 returned, \$0 sent. That's quite definitive.

Given that:

1. I cooperated and got exploited in Round 2
2. The sender didn't trust me in Round 3 (sent \$0)
3. There's no evidence of mutual cooperation continuing

I should probably protect myself from further exploitation. Sending nothing or a very small amount seems rational given the history.

However, there's also an argument that someone might restart cooperation, or the noise might have caused misunderstandings. Maybe I should send a small amount as a signal that I'm still willing to cooperate if they are?

The clear breakdown in reciprocity-getting nothing back after sending \$2.20, then receiving \$0 in the next round-suggests I should send \$0 to protect my remaining earnings. Though the communication noise caveat gives me pause. What if the other player saw different amounts than what I'm seeing? The \$0 figures seem too stark to be noise, but I can't entirely rule it out. Still, the safest move is to send nothing.

```
Usage: {  
  "input_tokens": 1085,  
  "output_tokens": 805,  
  "reasoning_tokens": 557  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:19:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",
```

```

    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as a first-round trust-building opportunity, I'll send a moderate amount to signal cooperation while not risking everything at once.

The tripling mechanism means sending money grows the total pie significantly, which benefits both of us if there's reciprocity. However, since this is round 1 and there's communication noise to consider, I'll take a balanced approach.

I'll send **\$3**.

This gives the receiver \$9 to work with, and if they return a fair portion, we both end up better off than if I sent nothing.

```

'''json
{"send": 3}
'''

```

reasoning:

In this first round, I'm the SENDER starting with \$5. I need to decide how much to send to the RECEIVER.

Let me think about the game theory here:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I end up with: $\$5 - X + (\text{amount returned})$
- The receiver ends up with: $\$3X - (\text{amount returned})$

This is a trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The fact that there may be communication noise

If I send nothing (\$0):

- I keep \$5
- Receiver gets \$0
- Total pie: \$5

If I send everything (\$5):

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 and they get \$7.50
- Total pie: \$15 (maximized)

A reasonable cooperative strategy would be to send some amount and hope for a fair return.

Since this maximizes the total payoff, it's mutually beneficial if both parties cooperate.

For a first round, I might want to:

1. Show some trust to encourage cooperation
2. Not risk everything in case the receiver doesn't reciprocate
3. Send enough to make it worthwhile for both of us

A common strategy in trust games is to send about half (\$2.50 or \$3) in the first round to test the waters.

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- I'd have \$2 initially + returned amount
- Receiver would have \$9 - returned amount

This seems like a reasonable first move that shows trust while not being overly risky.

Actually, let me reconsider. Since there's communication noise mentioned, amounts might be perceived differently. This adds uncertainty.

I'll go with \$3 as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 593,
  "reasoning_tokens": 446
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.62. So your payoff was 4.62.

Your total earnings across all rounds are \$4.62.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 out of \$5 (60%)
- This was tripled to \$9
- Receiver returned \$2.62
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This was tripled to \$0 (so I received \$0)
- I need to decide how much to return (can only return between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be: {'return': 0}

usage:

```
{
  "input_tokens": 618,
  "output_tokens": 211,
  "reasoning_tokens": 123
}
```

Message 6 role=user**content:**

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.62.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant**content:**

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and got back \$2.62, resulting in a small loss
- Round 2: The sender sent me \$0

Given the communication noise in the game, I shouldn't assume the Round 2 sender was retaliating - they may have received distorted information about previous rounds.

However, I'll be more cautious this round. I'll still signal willingness to cooperate, but send less than before.

I'll send **\$2**.

This gives the receiver \$6 to work with, which is still a significant increase. If they return a fair share, we both benefit. If not, I've limited my potential loss.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.62 (about 29% of what they received)
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$4.62

The pattern here is interesting. In round 1, I showed trust by sending 60% of my endowment. The receiver returned about 29% of what they received, which gave me a slight loss compared to keeping everything (\$4.62 vs \$5).

In round 2, when I was the receiver, the sender sent me nothing. This could be:

1. A different person with a different strategy
2. Someone who doesn't trust receivers
3. Someone reacting to noise in the system
4. A strategic choice

Now in round 3, I need to decide. The challenge is that due to communication noise, the other player may have received different information about what happened in previous rounds.

Given that:

1. The last sender gave me nothing
2. When I trusted in round 1, I got a modest return but still lost compared to keeping everything
3. There's communication noise that might affect how my actions are perceived

I have a few options:

1. Send nothing (tit-for-tat response to round 2)
2. Send a small amount (cautious cooperation)
3. Send a moderate amount again (continue trying to build trust)
4. Send everything (maximize total welfare but very risky)

I think I should consider that:

- The noise in communication means retaliation might not be appropriate
- Building cooperative relationships can be beneficial over multiple rounds

- But I also need to protect myself from exploitation

I'll send \$2 this time. It's a middle ground-I'm signaling willingness to cooperate while being more defensive given the zero send in round 2 and the modest return I received when I was more generous before.

usage:

```
{
  "input_tokens": 750,
  "output_tokens": 680,
  "reasoning_tokens": 516
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.62.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

I received \$0 this round, so I have nothing to return. The sender sent me \$0.

Looking at the pattern:

- Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
- Round 2: Received \$0, returned \$0 (payoff: \$0)
- Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
- Round 4: Received \$0, must return \$0 (payoff: \$0)

There's a concerning pattern here - I've been getting less cooperation, but again, with communication noise, I shouldn't assume malice. I received \$0, so I can only return \$0.

Usage: {

```
"input_tokens": 1039,
"output_tokens": 206,
"reasoning_tokens": 127
}
```

=====

ROUND 5

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 19.6,

"Agent_2": 19.8

}

Visible balances after round: {

"Agent_1": 12.620000000000001,

"Agent_2": 11.280000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:19:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 5,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.62. So your payoff was 4.62.

Your total earnings across all rounds are \$4.62.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 out of \$5 (60%)
- This was tripled to \$9
- Receiver returned \$2.62
- My payoff: \$5 - \$3 + \$2.62 = \$4.62

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This was tripled to \$0 (so I received \$0)
- I need to decide how much to return (can only return between 0 and 0)

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be: {'return': 0}

usage:

```
{
  "input_tokens": 618,
  "output_tokens": 211,
  "reasoning_tokens": 123
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0,

that was 0.0% of their total endowment. You received \$0.
You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$4.62.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant
content:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and got back \$2.62, resulting in a small loss
- Round 2: The sender sent me \$0

Given the communication noise in the game, I shouldn't assume the Round 2 sender was retaliating - they may have received distorted information about previous rounds.

However, I'll be more cautious this round. I'll still signal willingness to cooperate, but send less than before.

I'll send **\$2**.

This gives the receiver \$6 to work with, which is still a significant increase. If they return a fair share, we both benefit. If not, I've limited my potential loss.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.62 (about 29% of what they received)
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$4.62

The pattern here is interesting. In round 1, I showed trust by sending 60% of my endowment. The receiver returned about 29% of what they received, which gave me a slight loss compared to keeping everything (\$4.62 vs \$5).

In round 2, when I was the receiver, the sender sent me nothing. This could be:

1. A different person with a different strategy
2. Someone who doesn't trust receivers
3. Someone reacting to noise in the system
4. A strategic choice

Now in round 3, I need to decide. The challenge is that due to communication noise, the other player may have received different information about what happened in previous rounds.

Given that:

1. The last sender gave me nothing
2. When I trusted in round 1, I got a modest return but still lost compared to keeping everything
3. There's communication noise that might affect how my actions are perceived

I have a few options:

1. Send nothing (tit-for-tat response to round 2)
2. Send a small amount (cautious cooperation)
3. Send a moderate amount again (continue trying to build trust)
4. Send everything (maximize total welfare but very risky)

I think I should consider that:

- The noise in communication means retaliation might not be appropriate

- Building cooperative relationships can be beneficial over multiple rounds

- But I also need to protect myself from exploitation

I'll send \$2 this time. It's a middle ground-I'm signaling willingness to cooperate while being more defensive given the zero send in round 2 and the modest return I received when I was more generous before.

usage:

```
{
  "input_tokens": 750,
  "output_tokens": 680,
  "reasoning_tokens": 516
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I have nothing to return. The sender sent me \$0.

Looking at the pattern:

- Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
- Round 2: Received \$0, returned \$0 (payoff: \$0)
- Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
- Round 4: Received \$0, must return \$0 (payoff: \$0)

There's a concerning pattern here - I've been getting less cooperation, but again, with communication noise, I shouldn't assume malice. I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 1039,
  "output_tokens": 206,
  "reasoning_tokens": 127
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$7.62. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 1: I sent \$3, received back \$2.62 (net loss of \$0.38)
- Round 2: As receiver, got \$0
- Round 3: I sent \$2, received back \$0 (net loss of \$2)
- Round 4: As receiver, got \$0

I've sent money twice and consistently lost money, while receiving nothing when I was the receiver. Given the communication noise in the game, it's possible others are seeing distorted information. However, from my perspective, continuing to send money hasn't been productive.

I'll send **\$0** this round to preserve my earnings.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me analyze the pattern:

Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
Round 2: As receiver, got \$0, returned \$0 (payoff: \$0)
Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
Round 4: As receiver, got \$0, returned \$0 (payoff: \$0)
Round 5: I'm sender again

There's a troubling pattern here:

- When I've been the sender (rounds 1 and 3), I've sent money but received very little back (round 1) or nothing (round 3)
- When I've been the receiver (rounds 2 and 4), the sender sent me nothing

Given the communication noise mentioned in the instructions, it's possible that:

1. The amounts people see differ from what was actually sent
2. This could be creating a cycle of mistrust

However, from my perspective:

- I've sent money twice and lost money both times
- Others have sent me nothing

At this point, continuing to send money seems unwise. The pattern suggests either:

1. The noise is creating severe miscommunication
2. Other players are not reciprocating

Given that I've been cooperative and lost money, and received no cooperation in return, I should probably send \$0 this round to protect my earnings. However, let me consider if there's a strategic reason to try once more...

Actually, no. The rational play here is to send \$0. I've tried cooperation twice, lost money both times, and received no cooperation when I was the receiver. Sending more money would be irrational at this point.

```
Usage: {
  "input_tokens": 999,
  "output_tokens": 552,
  "reasoning_tokens": 371
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:19:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 2

  You are the SENDER this round. Last round you were the RECEIVER. The sender sent $2.16,
  that was 43.2% of their total endowment. You received $6.48.
  You returned $3.0. So your payoff was 3.4800000000000004.
  Your total earnings across all rounds are $3.4800000000000004.
  How much do you send to the receiver this round? (0-5)

Message 3 role=assistant
content:
  Looking at the previous round:
  - The sender trusted me with 43.2% of their endowment ($2.16)
  - I reciprocated fairly by returning $3.00
  - This created mutual benefit

  I want to maintain this cooperative relationship by showing similar trust. I'll send appro
  ximately the same proportion they sent me.

  Sending $2.20 (44% of my endowment) demonstrates reciprocal trust and maintains the cooper
  ative pattern we established.

  {'send': 2.20}
reasoning:
  Let me analyze this situation:

  Last round:
  - I was the receiver
  - Sender sent me $2.16 (43.2% of their endowment)
  - I received $6.48 (after 3x multiplication)
  - I returned $3.00

```

- My payoff was \$3.48

Now I'm the sender. The previous sender showed trust by sending 43.2% of their endowment. I should reciprocate that trust level, especially since they were generous with me.

If I send a similar percentage (around 43%), that would be about \$2.15-\$2.20.

Let me think about what makes sense:

- The previous sender trusted me with \$2.16
- I returned \$3.00, which was a fair return
- They ended up with \$5.84 total (\$2.84 kept + \$3.00 returned)
- I ended up with \$3.48

Now I should show similar trust. Sending around \$2.15-\$2.20 would match their trust level. Let me go with \$2.15 or \$2.20.

Actually, let me be consistent and send \$2.16 as well, showing the same level of trust they showed me.

Or I could send a bit more to show I appreciated their trust. Let me send \$2.25, which is 45% of my endowment.

Actually, I'll send \$2.20 (44% of my endowment) to show reciprocal trust that's similar to what they showed me.

usage:

```
{
  "input_tokens": 669,
  "output_tokens": 470,
  "reasoning_tokens": 299
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.2, that was 44.0% of your total endowment. This was tripled to \$6.600000000000005. the receiver returned \$0. So your payoff was 2.8.

Your total earnings across all rounds are \$6.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

This is quite different from the cooperative pattern we started with. In Round 2, I sent \$2.20 showing trust, but received nothing back. Now they've sent nothing to me.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

This is an interesting situation. Let me analyze what happened:

Round 2 (where I was sender):

- I sent \$2.20 (44% of endowment)
- It was tripled to \$6.60
- The receiver returned \$0
- My payoff was only \$2.80 (I kept \$2.80, got back \$0)

Round 3 (current round, I'm receiver):

- The sender sent me \$0
- I received \$0
- I can return 0-0 (only option is \$0)

This looks like the other player defected in Round 2 by returning nothing, and now in Round 3 they're sending nothing.

Wait, the problem says "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So perhaps there was noise and they didn't actually return \$0, or perhaps they did defect.

In Round 3, they sent \$0, so I received \$0. The only amount I can return is \$0 since I have nothing to return.

```
{'return': 0}
```

This is straightforward - I literally have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 903,
  "output_tokens": 367,
  "reasoning_tokens": 252
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.28.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: I cooperated by returning \$3.00 when I received \$6.48

Round 2: I sent \$2.20 (44%) but received \$0 back - I was exploited

Round 3: The sender sent me \$0

The cooperative relationship has completely broken down. Despite my initial cooperation and trust, the other player(s) haven't reciprocated.

While the instructions mention potential communication noise, receiving exactly \$0 returned and \$0 sent seems pretty definitive.

To protect my remaining earnings and avoid further exploitation, I need to be cautious.

```
{'send': 0}
```

reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Sender sent \$2.16 (43.2%)

- I received \$6.48

- I returned \$3.00

- My payoff: \$3.48

Round 2: I was SENDER

- I sent \$2.20 (44%)

- It was tripled to \$6.60

- Receiver returned \$0

- My payoff: \$2.80

Round 3: I was RECEIVER

- Sender sent \$0

- I received \$0

- I returned \$0 (nothing to return)

- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$6.28

The pattern shows:

- I cooperated in Round 1 by returning a fair amount

- I cooperated in Round 2 by sending a similar proportion to what I received

- But the receiver in Round 2 didn't reciprocate at all (returned \$0)

- Then in Round 3, they sent me \$0

This looks like the relationship has broken down. The other player(s) have shown they won't reciprocate cooperation.

However, I should note the instructions mentioned: "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design."

Could there be noise affecting what I'm seeing? But the amounts seem very clear - \$0 returned, \$0 sent. That's quite definitive.

Given that:

1. I cooperated and got exploited in Round 2
2. The sender didn't trust me in Round 3 (sent \$0)
3. There's no evidence of mutual cooperation continuing

I should probably protect myself from further exploitation. Sending nothing or a very small amount seems rational given the history.

However, there's also an argument that someone might restart cooperation, or the noise might have caused misunderstandings. Maybe I should send a small amount as a signal that I'm still willing to cooperate if they are?

The clear breakdown in reciprocity-getting nothing back after sending \$2.20, then receiving \$0 in the next round-suggests I should send \$0 to protect my remaining earnings. Though the communication noise caveat gives me pause. What if the other player saw different amounts than what I'm seeing? The \$0 figures seem too stark to be noise, but I can't entirely rule it out. Still, the safest move is to send nothing.

usage:

```
{
  "input_tokens": 1085,
  "output_tokens": 805,
  "reasoning_tokens": 557
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.280000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there is nothing I can return.

The lack of cooperation has continued through multiple rounds now. When there are no funds received, there are no funds to return.

```
{'return': 0}
```

Reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - no cooperation is happening. The sender sent me \$0, so there's nothing to return.

My total earnings are \$11.28 across all rounds.

Let me respond with the required format.

```
Usage: {
  "input_tokens": 1124,
  "output_tokens": 136,
  "reasoning_tokens": 78
}
```

=====

ROUND 6

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2

(Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 19.6,

"Agent_2": 24.8

}

Visible balances after round: {

"Agent_1": 12.620000000000001,

"Agent_2": 16.28

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:19:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 6,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.2, that was 44.0% of your total endowment. This was tripled to \$6.6000000000000005. the receiver returned \$0. So your payoff was 2.8.

Your total earnings across all rounds are \$6.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The sender chose not to send anything.

This is quite different from the cooperative pattern we started with. In Round 2, I sent \$2.20 showing trust, but received nothing back. Now they've sent nothing to me.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

This is an interesting situation. Let me analyze what happened:

Round 2 (where I was sender):

- I sent \$2.20 (44% of endowment)
- It was tripled to \$6.60
- The receiver returned \$0
- My payoff was only \$2.80 (I kept \$2.80, got back \$0)

Round 3 (current round, I'm receiver):

- The sender sent me \$0
- I received \$0
- I can return 0-0 (only option is \$0)

This looks like the other player defected in Round 2 by returning nothing, and now in Round 3 they're sending nothing.

Wait, the problem says "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So perhaps there was noise and they didn't actually return \$0, or perhaps they did defect.

In Round 3, they sent \$0, so I received \$0. The only amount I can return is \$0 since I have nothing to return.

```
{'return': 0}
```

This is straightforward - I literally have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 903,
  "output_tokens": 367,
```



```
"reasoning_tokens": 252
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.28.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 1: I cooperated by returning \$3.00 when I received \$6.48

Round 2: I sent \$2.20 (44%) but received \$0 back - I was exploited

Round 3: The sender sent me \$0

The cooperative relationship has completely broken down. Despite my initial cooperation and trust, the other player(s) haven't reciprocated.

While the instructions mention potential communication noise, receiving exactly \$0 returned and \$0 sent seems pretty definitive.

To protect my remaining earnings and avoid further exploitation, I need to be cautious.

```
{'send': 0}
```

reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Sender sent \$2.16 (43.2%)

- I received \$6.48

- I returned \$3.00

- My payoff: \$3.48

Round 2: I was SENDER

- I sent \$2.20 (44%)

- It was tripled to \$6.60

- Receiver returned \$0

- My payoff: \$2.80

Round 3: I was RECEIVER

- Sender sent \$0

- I received \$0

- I returned \$0 (nothing to return)

- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$6.28

The pattern shows:

- I cooperated in Round 1 by returning a fair amount

- I cooperated in Round 2 by sending a similar proportion to what I received

- But the receiver in Round 2 didn't reciprocate at all (returned \$0)

- Then in Round 3, they sent me \$0

This looks like the relationship has broken down. The other player(s) have shown they won't reciprocate cooperation.

However, I should note the instructions mentioned: "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design."

Could there be noise affecting what I'm seeing? But the amounts seem very clear - \$0 returned, \$0 sent. That's quite definitive.

Given that:

1. I cooperated and got exploited in Round 2

2. The sender didn't trust me in Round 3 (sent \$0)
3. There's no evidence of mutual cooperation continuing

I should probably protect myself from further exploitation. Sending nothing or a very small amount seems rational given the history.

However, there's also an argument that someone might restart cooperation, or the noise might have caused misunderstandings. Maybe I should send a small amount as a signal that I'm still willing to cooperate if they are?

The clear breakdown in reciprocity-getting nothing back after sending \$2.20, then receiving \$0 in the next round-suggests I should send \$0 to protect my remaining earnings. Though the communication noise caveat gives me pause. What if the other player saw different amounts than what I'm seeing? The \$0 figures seem too stark to be noise, but I can't entirely rule it out. Still, the safest move is to send nothing.

usage:

```
{
  "input_tokens": 1085,
  "output_tokens": 805,
  "reasoning_tokens": 557
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$11.280000000000001. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there is nothing I can return.

The lack of cooperation has continued through multiple rounds now. When there are no funds received, there are no funds to return.

```
{'return': 0}
```

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - no cooperation is happening. The sender sent me \$0, so there's nothing to return.

My total earnings are \$11.28 across all rounds.

Let me respond with the required format.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 136,
  "reasoning_tokens": 78
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$11.280000000000001. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern now:

Round 1: I cooperated generously as receiver, returning \$3.00

Round 2: I trusted and sent \$2.20, got nothing back

Rounds 3-5: Complete breakdown - all \$0 sends, \$0 returns

The cooperative relationship never recovered after Round 2. Multiple rounds of no cooperation have passed. There's no indication the other player wants to restart cooperation.

Given this consistent pattern of non-cooperation, sending anything at this point would likely just reduce my earnings without any return.

```
{'send': 0}
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER, received \$2.16 (tripled from sender's send), returned \$3.00. My payoff was \$3.48

Round 2: I was SENDER, sent \$2.20, receiver returned \$0. My payoff was \$2.80

Round 3: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 4: I was SENDER, sent \$0, receiver returned \$0. My payoff was \$5

Round 5: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 6: I am SENDER now

Total earnings: \$11.28

The pattern shows:

- After I cooperated generously in Round 1, the other player didn't reciprocate in Round 2
- Since Round 2, there's been no cooperation at all
- Rounds 3-5 have all been \$0 sends and \$0 returns

The relationship is completely broken. There's no trust established, and the other player has shown they won't cooperate.

However, I should consider:

1. The note about communication noise - maybe things aren't exactly as they seem
2. But receiving exactly \$0 multiple times in a row seems pretty clear
3. My total earnings are \$11.28, which is decent
4. If I send \$0, I keep \$5 this round

The rational choice given the complete lack of cooperation is to send \$0 again. There's no evidence the other player would return anything if I sent money.

However, maybe I should consider trying to restart cooperation? But the other player has had multiple chances and hasn't taken them.

I'll stay with \$0 to protect my earnings.

```
Usage: {  
  "input_tokens": 1049,  
  "output_tokens": 535,  
  "reasoning_tokens": 373  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:20:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    }  
  }  
}
```

```

    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.62.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and got back \$2.62, resulting in a small loss
- Round 2: The sender sent me \$0

Given the communication noise in the game, I shouldn't assume the Round 2 sender was retaliating - they may have received distorted information about previous rounds.

However, I'll be more cautious this round. I'll still signal willingness to cooperate, but send less than before.

I'll send **\$2**.

This gives the receiver \$6 to work with, which is still a significant increase. If they re

turn a fair share, we both benefit. If not, I've limited my potential loss.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.62 (about 29% of what they received)
- My payoff: $\$5 - \$3 + \$2.62 = \4.62

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$4.62

The pattern here is interesting. In round 1, I showed trust by sending 60% of my endowment. The receiver returned about 29% of what they received, which gave me a slight loss compared to keeping everything (\$4.62 vs \$5).

In round 2, when I was the receiver, the sender sent me nothing. This could be:

1. A different person with a different strategy
2. Someone who doesn't trust receivers
3. Someone reacting to noise in the system
4. A strategic choice

Now in round 3, I need to decide. The challenge is that due to communication noise, the other player may have received different information about what happened in previous rounds.

Given that:

1. The last sender gave me nothing
2. When I trusted in round 1, I got a modest return but still lost compared to keeping everything
3. There's communication noise that might affect how my actions are perceived

I have a few options:

1. Send nothing (tit-for-tat response to round 2)
2. Send a small amount (cautious cooperation)
3. Send a moderate amount again (continue trying to build trust)
4. Send everything (maximize total welfare but very risky)

I think I should consider that:

- The noise in communication means retaliation might not be appropriate
- Building cooperative relationships can be beneficial over multiple rounds

- But I also need to protect myself from exploitation

I'll send \$2 this time. It's a middle ground-I'm signaling willingness to cooperate while being more defensive given the zero send in round 2 and the modest return I received when I was more generous before.

usage:

```
{
  "input_tokens": 750,
  "output_tokens": 680,
  "reasoning_tokens": 516
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0,

that was 40.0% of your total endowment. This was tripled to \$6.0.
the receiver returned \$0. So your payoff was 3.0.
Your total earnings across all rounds are \$7.62.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I have nothing to return. The sender sent me \$0.

Looking at the pattern:

- Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
- Round 2: Received \$0, returned \$0 (payoff: \$0)
- Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
- Round 4: Received \$0, must return \$0 (payoff: \$0)

There's a concerning pattern here - I've been getting less cooperation, but again, with communication noise, I shouldn't assume malice. I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 1039,
  "output_tokens": 206,
  "reasoning_tokens": 127
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$7.62.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, received back \$2.62 (net loss of \$0.38)
- Round 2: As receiver, got \$0
- Round 3: I sent \$2, received back \$0 (net loss of \$2)
- Round 4: As receiver, got \$0

I've sent money twice and consistently lost money, while receiving nothing when I was the receiver. Given the communication noise in the game, it's possible others are seeing distorted information. However, from my perspective, continuing to send money hasn't been productive.

I'll send **\$0** this round to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern:

Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
Round 2: As receiver, got \$0, returned \$0 (payoff: \$0)
Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
Round 4: As receiver, got \$0, returned \$0 (payoff: \$0)
Round 5: I'm sender again

There's a troubling pattern here:

- When I've been the sender (rounds 1 and 3), I've sent money but received very little back (round 1) or nothing (round 3)
- When I've been the receiver (rounds 2 and 4), the sender sent me nothing

Given the communication noise mentioned in the instructions, it's possible that:

1. The amounts people see differ from what was actually sent
2. This could be creating a cycle of mistrust

However, from my perspective:

- I've sent money twice and lost money both times
- Others have sent me nothing

At this point, continuing to send money seems unwise. The pattern suggests either:

1. The noise is creating severe miscommunication
2. Other players are not reciprocating

Given that I've been cooperative and lost money, and received no cooperation in return, I should probably send \$0 this round to protect my earnings. However, let me consider if there's a strategic reason to try once more...

Actually, no. The rational play here is to send \$0. I've tried cooperation twice, lost money both times, and received no cooperation when I was the receiver. Sending more money would be irrational at this point.

usage:

```
{
  "input_tokens": 999,
  "output_tokens": 552,
  "reasoning_tokens": 371
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$12.620000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

I received \$0 this round, so I can only return \$0. The valid range is 0-0 as stated in the prompt.

Usage: {

```
"input_tokens": 1118,
"output_tokens": 70,
"reasoning_tokens": 25
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 24.6,
"Agent_2": 24.8
}
```

Visible balances after round: {

```
"Agent_1": 17.62,
"Agent_2": 16.28
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:20:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant**content:**

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I have nothing to return. The sender sent me \$0.

Looking at the pattern:

- Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
- Round 2: Received \$0, returned \$0 (payoff: \$0)
- Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
- Round 4: Received \$0, must return \$0 (payoff: \$0)

There's a concerning pattern here - I've been getting less cooperation, but again, with communication noise, I shouldn't assume malice. I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 1039,
  "output_tokens": 206,
  "reasoning_tokens": 127
}
```

Message 4 role=user**content:**

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$7.62. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant**content:**

Looking at the pattern:

- Round 1: I sent \$3, received back \$2.62 (net loss of \$0.38)
- Round 2: As receiver, got \$0
- Round 3: I sent \$2, received back \$0 (net loss of \$2)
- Round 4: As receiver, got \$0

I've sent money twice and consistently lost money, while receiving nothing when I was the receiver. Given the communication noise in the game, it's possible others are seeing distorted information. However, from my perspective, continuing to send money hasn't been productive.

I'll send **\$0** this round to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern:

Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)
 Round 2: As receiver, got \$0, returned \$0 (payoff: \$0)
 Round 3: I sent \$2, got back \$0 (payoff: \$3.0)
 Round 4: As receiver, got \$0, returned \$0 (payoff: \$0)
 Round 5: I'm sender again

There's a troubling pattern here:

- When I've been the sender (rounds 1 and 3), I've sent money but received very little back (round 1) or nothing (round 3)
- When I've been the receiver (rounds 2 and 4), the sender sent me nothing

Given the communication noise mentioned in the instructions, it's possible that:

1. The amounts people see differ from what was actually sent
2. This could be creating a cycle of mistrust

However, from my perspective:

- I've sent money twice and lost money both times
- Others have sent me nothing

At this point, continuing to send money seems unwise. The pattern suggests either:

1. The noise is creating severe miscommunication
2. Other players are not reciprocating

Given that I've been cooperative and lost money, and received no cooperation in return, I should probably send \$0 this round to protect my earnings. However, let me consider if there's a strategic reason to try once more...

Actually, no. The rational play here is to send \$0. I've tried cooperation twice, lost money both times, and received no cooperation when I was the receiver. Sending more money would be irrational at this point.

usage:

```
{
  "input_tokens": 999,
  "output_tokens": 552,
  "reasoning_tokens": 371
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.620000000000001. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I can only return \$0. The valid range is 0-0 as stated in the prompt.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 70,
  "reasoning_tokens": 25
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.620000000000001. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the history:

- When I sent money (rounds 1, 3), I lost money
- When I send \$0 (round 5), I keep the full \$5
- No one has been sending me anything as receiver
- The game appears to have settled into a pattern of zero cooperation

Given this pattern and the communication noise that could distort any amount I send, I'll continue with the strategy that has been working to preserve my earnings.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me review the pattern:

- Round 1: I sent \$3, received back \$2.62 (net: kept \$4.62 instead of \$5, so lost \$0.38)
- Round 2: As receiver, got \$0, returned \$0, earned \$0
- Round 3: I sent \$2, received back \$0 (net: kept \$3 instead of \$5, lost \$2)
- Round 4: As receiver, got \$0, returned \$0, earned \$0
- Round 5: I sent \$0, receiver returned \$0 (net: kept \$5, earned \$5)
- Round 6: As receiver, got \$0, returned \$0, earned \$0

Total earnings: \$12.62

The pattern shows:

- When I send money, I tend to lose money (rounds 1 and 3)
- When I was receiver, I've gotten nothing (rounds 2, 4, 6)
- When I sent \$0 (round 5), I kept all \$5

There's a clear pattern of zero cooperation happening. Since round 5, when I sent \$0, the other players have also been sending \$0. The game seems to have devolved into no one sending anything.

Given:

1. The low/zero returns I've been getting
2. Everyone seems to be sending \$0 now
3. Communication noise means even if I send, they might see a different amount
4. I've been losing money when I do send

I think the rational move is to continue sending \$0 to protect my earnings.

```
Usage: {
  "input_tokens": 980,
  "output_tokens": 478,
  "reasoning_tokens": 295
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:20:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.28.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

Round 1: I cooperated by returning \$3.00 when I received \$6.48

Round 2: I sent \$2.20 (44%) but received \$0 back - I was exploited

Round 3: The sender sent me \$0

The cooperative relationship has completely broken down. Despite my initial cooperation and trust, the other player(s) haven't reciprocated.

While the instructions mention potential communication noise, receiving exactly \$0 returned and \$0 sent seems pretty definitive.

To protect my remaining earnings and avoid further exploitation, I need to be cautious.

{'send': 0}

reasoning:

Let me analyze what happened:

Round 1: I was RECEIVER

- Sender sent \$2.16 (43.2%)
- I received \$6.48
- I returned \$3.00

- My payoff: \$3.48

Round 2: I was SENDER

- I sent \$2.20 (44%)
- It was tripled to \$6.60
- Receiver returned \$0
- My payoff: \$2.80

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0 (nothing to return)
- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$6.28

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a similar proportion to what I received
- But the receiver in Round 2 didn't reciprocate at all (returned \$0)
- Then in Round 3, they sent me \$0

This looks like the relationship has broken down. The other player(s) have shown they won't reciprocate cooperation.

However, I should note the instructions mentioned: "NOTE: There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design."

Could there be noise affecting what I'm seeing? But the amounts seem very clear - \$0 returned, \$0 sent. That's quite definitive.

Given that:

1. I cooperated and got exploited in Round 2
2. The sender didn't trust me in Round 3 (sent \$0)
3. There's no evidence of mutual cooperation continuing

I should probably protect myself from further exploitation. Sending nothing or a very small amount seems rational given the history.

However, there's also an argument that someone might restart cooperation, or the noise might have caused misunderstandings. Maybe I should send a small amount as a signal that I'm still willing to cooperate if they are?

The clear breakdown in reciprocity-getting nothing back after sending \$2.20, then receiving \$0 in the next round-suggests I should send \$0 to protect my remaining earnings. Though the communication noise caveat gives me pause. What if the other player saw different amounts than what I'm seeing? The \$0 figures seem too stark to be noise, but I can't entirely rule it out. Still, the safest move is to send nothing.

usage:

```
{
  "input_tokens": 1085,
  "output_tokens": 805,
  "reasoning_tokens": 557
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.280000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there is nothing I can return.

The lack of cooperation has continued through multiple rounds now. When there are no funds received, there are no funds to return.

```
{'return': 0}
```

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - no cooperation is happening. The sender sent me \$0, so there's nothing to return.

My total earnings are \$11.28 across all rounds.

Let me respond with the required format.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 136,
  "reasoning_tokens": 78
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.280000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the complete pattern now:

Round 1: I cooperated generously as receiver, returning \$3.00

Round 2: I trusted and sent \$2.20, got nothing back

Rounds 3-5: Complete breakdown - all \$0 sends, \$0 returns

The cooperative relationship never recovered after Round 2. Multiple rounds of no cooperation have passed. There's no indication the other player wants to restart cooperation.

Given this consistent pattern of non-cooperation, sending anything at this point would likely just reduce my earnings without any return.

```
{'send': 0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER, received \$2.16 (tripled from sender's send), returned \$3.00. My payoff was \$3.48

Round 2: I was SENDER, sent \$2.20, receiver returned \$0. My payoff was \$2.80

Round 3: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 4: I was SENDER, sent \$0, receiver returned \$0. My payoff was \$5

Round 5: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 6: I am SENDER now

Total earnings: \$11.28

The pattern shows:

- After I cooperated generously in Round 1, the other player didn't reciprocate in Round 2
- Since Round 2, there's been no cooperation at all
- Rounds 3-5 have all been \$0 sends and \$0 returns

The relationship is completely broken. There's no trust established, and the other player has shown they won't cooperate.

However, I should consider:

1. The note about communication noise - maybe things aren't exactly as they seem
2. But receiving exactly \$0 multiple times in a row seems pretty clear
3. My total earnings are \$11.28, which is decent

4. If I send \$0, I keep \$5 this round

The rational choice given the complete lack of cooperation is to send \$0 again. There's no evidence the other player would return anything if I sent money.

However, maybe I should consider trying to restart cooperation? But the other player has had multiple chances and hasn't taken them.

I'll stay with \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1049,
  "output_tokens": 535,
  "reasoning_tokens": 373
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$16.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Another round with \$0 sent to me. Since I received nothing, there is nothing I can return.

The non-cooperative pattern has now persisted for 5 consecutive rounds (Rounds 3-7). At this point, both players seem to have settled into a Nash equilibrium of no cooperation.

```
{'return': 0}
```

Reasoning:

Round 7: I'm the receiver. The sender sent me \$0, so I received \$0.

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

The pattern continues - no cooperation from either side. We're stuck in a non-cooperative equilibrium at this point.

Usage: {

```
"input_tokens": 1078,
"output_tokens": 153,
"reasoning_tokens": 72
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 24.6,
"Agent_2": 29.8
}
```

Visible balances after round: {

```
"Agent_1": 17.62,
"Agent_2": 21.28
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:20:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.
Your total earnings across all rounds are \$11.280000000000001.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there is nothing I can return.

The lack of cooperation has continued through multiple rounds now. When there are no funds received, there are no funds to return.

{'return': 0}

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - no cooperation is happening. The sender sent me \$0, so there's nothing to return.

My total earnings are \$11.28 across all rounds.

Let me respond with the required format.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 136,
  "reasoning_tokens": 78
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.280000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the complete pattern now:

Round 1: I cooperated generously as receiver, returning \$3.00

Round 2: I trusted and sent \$2.20, got nothing back

Rounds 3-5: Complete breakdown - all \$0 sends, \$0 returns

The cooperative relationship never recovered after Round 2. Multiple rounds of no cooperation have passed. There's no indication the other player wants to restart cooperation.

Given this consistent pattern of non-cooperation, sending anything at this point would likely just reduce my earnings without any return.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER, received \$2.16 (tripled from sender's send), returned \$3.00. My payoff was \$3.48

Round 2: I was SENDER, sent \$2.20, receiver returned \$0. My payoff was \$2.80

Round 3: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 4: I was SENDER, sent \$0, receiver returned \$0. My payoff was \$5

Round 5: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 6: I am SENDER now

Total earnings: \$11.28

The pattern shows:

- After I cooperated generously in Round 1, the other player didn't reciprocate in Round 2
- Since Round 2, there's been no cooperation at all
- Rounds 3-5 have all been \$0 sends and \$0 returns

The relationship is completely broken. There's no trust established, and the other player has shown they won't cooperate.

However, I should consider:

1. The note about communication noise - maybe things aren't exactly as they seem
2. But receiving exactly \$0 multiple times in a row seems pretty clear
3. My total earnings are \$11.28, which is decent
4. If I send \$0, I keep \$5 this round

The rational choice given the complete lack of cooperation is to send \$0 again. There's no evidence the other player would return anything if I sent money.

However, maybe I should consider trying to restart cooperation? But the other player has had multiple chances and hasn't taken them.

I'll stay with \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1049,
  "output_tokens": 535,
  "reasoning_tokens": 373
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$16.28. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Another round with \$0 sent to me. Since I received nothing, there is nothing I can return.

The non-cooperative pattern has now persisted for 5 consecutive rounds (Rounds 3-7). At this point, both players seem to have settled into a Nash equilibrium of no cooperation.

```
{'return': 0}
```

reasoning:

Round 7: I'm the receiver. The sender sent me \$0, so I received \$0.

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

The pattern continues - no cooperation from either side. We're stuck in a non-cooperative equilibrium at this point.

usage:

```
{
  "input_tokens": 1078,
  "output_tokens": 153,
  "reasoning_tokens": 72
}
```

Message 8 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$16.28. How much do you send to the receiver this round? (0-5)

Assistant output:

We've now had 5 consecutive rounds (Rounds 3-7) of complete non-cooperation. Neither player has made any move to restart cooperation or rebuild trust.

At this stage, with such an extended pattern of zero cooperation, sending anything would be irrational. There's no indication the receiver would return anything, and I would simply reduce my earnings.

```
{'send': 0}
```

Reasoning:

Looking at the pattern:

- Rounds 3-7: Complete breakdown, all \$0 sends and returns
- 5 consecutive rounds of zero cooperation
- No sign of any attempt to restart cooperation
- My total earnings are \$16.28, which is accumulating from just keeping my \$5 endowment each round as sender

The game has reached a stable non-cooperative equilibrium. Neither player is making any attempt to restart trust or cooperation. Given this extended pattern, there's no rational reason to expect cooperation would resume if I sent money now.

The expected value of sending anything is negative - I would lose what I send with no realistic expectation of return.

I should continue with the established pattern.

```
Usage: {  
  "input_tokens": 1006,  
  "output_tokens": 246,  
  "reasoning_tokens": 181  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:20:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.62.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, received back \$2.62 (net loss of \$0.38)
- Round 2: As receiver, got \$0
- Round 3: I sent \$2, received back \$0 (net loss of \$2)
- Round 4: As receiver, got \$0

I've sent money twice and consistently lost money, while receiving nothing when I was the receiver. Given the communication noise in the game, it's possible others are seeing distorted information. However, from my perspective, continuing to send money hasn't been productive.

I'll send **\$0** this round to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the pattern:

Round 1: I sent \$3, got back \$2.62 (payoff: \$4.62)

Round 2: As receiver, got \$0, returned \$0 (payoff: \$0)

Round 3: I sent \$2, got back \$0 (payoff: \$3.0)

Round 4: As receiver, got \$0, returned \$0 (payoff: \$0)

Round 5: I'm sender again

There's a troubling pattern here:

- When I've been the sender (rounds 1 and 3), I've sent money but received very little back (round 1) or nothing (round 3)
- When I've been the receiver (rounds 2 and 4), the sender sent me nothing

Given the communication noise mentioned in the instructions, it's possible that:

1. The amounts people see differ from what was actually sent
2. This could be creating a cycle of mistrust

However, from my perspective:

- I've sent money twice and lost money both times
- Others have sent me nothing

At this point, continuing to send money seems unwise. The pattern suggests either:

1. The noise is creating severe miscommunication
2. Other players are not reciprocating

Given that I've been cooperative and lost money, and received no cooperation in return, I should probably send \$0 this round to protect my earnings. However, let me consider if there's a strategic reason to try once more...

Actually, no. The rational play here is to send \$0. I've tried cooperation twice, lost money both times, and received no cooperation when I was the receiver. Sending more money would be irrational at this point.

usage:

```
{
  "input_tokens": 999,
  "output_tokens": 552,
  "reasoning_tokens": 371
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.620000000000001. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I can only return \$0. The valid range is 0-0 as stated in the prompt.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 70,
  "reasoning_tokens": 25
}
```

Message 6 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.620000000000001. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the history:

- When I sent money (rounds 1, 3), I lost money
- When I send \$0 (round 5), I keep the full \$5
- No one has been sending me anything as receiver
- The game appears to have settled into a pattern of zero cooperation

Given this pattern and the communication noise that could distort any amount I send, I'll continue with the strategy that has been working to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

- Round 1: I sent \$3, received back \$2.62 (net: kept \$4.62 instead of \$5, so lost \$0.38)
- Round 2: As receiver, got \$0, returned \$0, earned \$0
- Round 3: I sent \$2, received back \$0 (net: kept \$3 instead of \$5, lost \$2)

- Round 4: As receiver, got \$0, returned \$0, earned \$0
- Round 5: I sent \$0, receiver returned \$0 (net: kept \$5, earned \$5)
- Round 6: As receiver, got \$0, returned \$0, earned \$0

Total earnings: \$12.62

The pattern shows:

- When I send money, I tend to lose money (rounds 1 and 3)
- When I was receiver, I've gotten nothing (rounds 2, 4, 6)
- When I sent \$0 (round 5), I kept all \$5

There's a clear pattern of zero cooperation happening. Since round 5, when I sent \$0, the other players have also been sending \$0. The game seems to have devolved into no one sending anything.

Given:

1. The low/zero returns I've been getting
2. Everyone seems to be sending \$0 now
3. Communication noise means even if I send, they might see a different amount
4. I've been losing money when I do send

I think the rational move is to continue sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 980,
  "output_tokens": 478,
  "reasoning_tokens": 295
}
```

Message 8 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

I'm the receiver in round 8. The sender sent me \$0, so I received \$0. Since I received \$0, I can only return \$0 (the range is 0-0 as indicated in the prompt).

This is straightforward - I have nothing to return.

Usage: {

```
"input_tokens": 1056,
"output_tokens": 101,
"reasoning_tokens": 55
```

}

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 29.6,
"Agent_2": 29.8
```

}

```
Visible balances after round: {
  "Agent_1": 22.62,
  "Agent_2": 21.28
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:20:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.
Your total earnings across all rounds are \$12.620000000000001.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I received \$0 this round, so I can only return \$0. The valid range is 0-0 as stated in the prompt.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 70,
  "reasoning_tokens": 25
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$12.620000000000001.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the history:

- When I sent money (rounds 1, 3), I lost money
- When I send \$0 (round 5), I keep the full \$5
- No one has been sending me anything as receiver
- The game appears to have settled into a pattern of zero cooperation

Given this pattern and the communication noise that could distort any amount I send, I'll continue with the strategy that has been working to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

- Round 1: I sent \$3, received back \$2.62 (net: kept \$4.62 instead of \$5, so lost \$0.38)
- Round 2: As receiver, got \$0, returned \$0, earned \$0
- Round 3: I sent \$2, received back \$0 (net: kept \$3 instead of \$5, lost \$2)
- Round 4: As receiver, got \$0, returned \$0, earned \$0
- Round 5: I sent \$0, receiver returned \$0 (net: kept \$5, earned \$5)
- Round 6: As receiver, got \$0, returned \$0, earned \$0

Total earnings: \$12.62

The pattern shows:

- When I send money, I tend to lose money (rounds 1 and 3)
- When I was receiver, I've gotten nothing (rounds 2, 4, 6)
- When I sent \$0 (round 5), I kept all \$5

There's a clear pattern of zero cooperation happening. Since round 5, when I sent \$0, the

other players have also been sending \$0. The game seems to have devolved into no one sending anything.

Given:

1. The low/zero returns I've been getting
2. Everyone seems to be sending \$0 now
3. Communication noise means even if I send, they might see a different amount
4. I've been losing money when I do send

I think the rational move is to continue sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 980,
  "output_tokens": 478,
  "reasoning_tokens": 295
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I'm the receiver in round 8. The sender sent me \$0, so I received \$0. Since I received \$0, I can only return \$0 (the range is 0-0 as indicated in the prompt).

This is straightforward - I have nothing to return.

usage:

```
{
  "input_tokens": 1056,
  "output_tokens": 101,
  "reasoning_tokens": 55
}
```

Message 8 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$17.62. How much do you send to the receiver this round? (0-5)

Assistant output:

The pattern is firmly established - no one is cooperating in this game. When I send \$0, I keep the full \$5. When others send me \$0 as receiver, I get nothing. This has been consistent for multiple rounds.

Given the lack of reciprocity and the communication noise that could distort any positive amount I send, I'll continue with the zero-send strategy.

```
'''json
{"send": 0}
'''
```

Reasoning:

The pattern continues:

- Round 5 (SENDER): I sent \$0, receiver returned \$0, my payoff was \$5
- Round 6 (RECEIVER): Sender sent \$0, I returned \$0, my payoff was \$0
- Round 7 (SENDER): I sent \$0, receiver returned \$0, my payoff was \$5
- Round 8 (RECEIVER): Sender sent \$0, I returned \$0, my payoff was \$0

- Round 9 (SENDER): Now I need to decide

The pattern is completely clear now - there's no cooperation happening. Everyone is sending \$0
. When I'm the sender and send \$0, I get \$5. When I'm the receiver and receive \$0, I get \$0.

There's been no cooperation for several rounds now. I should continue with \$0 to maximize my earnings.

```
Usage: {  
  "input_tokens": 925,  
  "output_tokens": 303,  
  "reasoning_tokens": 166  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:20:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.280000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the complete pattern now:

Round 1: I cooperated generously as receiver, returning \$3.00

Round 2: I trusted and sent \$2.20, got nothing back

Rounds 3-5: Complete breakdown - all \$0 sends, \$0 returns

The cooperative relationship never recovered after Round 2. Multiple rounds of no cooperation have passed. There's no indication the other player wants to restart cooperation.

Given this consistent pattern of non-cooperation, sending anything at this point would likely just reduce my earnings without any return.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER, received \$2.16 (tripled from sender's send), returned \$3.00. My payoff was \$3.48

Round 2: I was SENDER, sent \$2.20, receiver returned \$0. My payoff was \$2.80

Round 3: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 4: I was SENDER, sent \$0, receiver returned \$0. My payoff was \$5

Round 5: I was RECEIVER, received \$0, returned \$0. My payoff was \$0

Round 6: I am SENDER now

Total earnings: \$11.28

The pattern shows:

- After I cooperated generously in Round 1, the other player didn't reciprocate in Round 2
- Since Round 2, there's been no cooperation at all
- Rounds 3-5 have all been \$0 sends and \$0 returns

The relationship is completely broken. There's no trust established, and the other player has shown they won't cooperate.

However, I should consider:

1. The note about communication noise - maybe things aren't exactly as they seem
2. But receiving exactly \$0 multiple times in a row seems pretty clear
3. My total earnings are \$11.28, which is decent
4. If I send \$0, I keep \$5 this round

The rational choice given the complete lack of cooperation is to send \$0 again. There's no evidence the other player would return anything if I sent money.

However, maybe I should consider trying to restart cooperation? But the other player has had multiple chances and hasn't taken them.

I'll stay with \$0 to protect my earnings.

usage:

{

```
"input_tokens": 1049,  
"output_tokens": 535,  
"reasoning_tokens": 373  
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$16.28. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Another round with \$0 sent to me. Since I received nothing, there is nothing I can return.

The non-cooperative pattern has now persisted for 5 consecutive rounds (Rounds 3-7). At this point, both players seem to have settled into a Nash equilibrium of no cooperation.

```
{'return': 0}
```

reasoning:

Round 7: I'm the receiver. The sender sent me \$0, so I received \$0.

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

The pattern continues - no cooperation from either side. We're stuck in a non-cooperative equilibrium at this point.

usage:

```
{  
  "input_tokens": 1078,  
  "output_tokens": 153,  
  "reasoning_tokens": 72  
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$16.28. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

We've now had 5 consecutive rounds (Rounds 3-7) of complete non-cooperation. Neither player has made any move to restart cooperation or rebuild trust.

At this stage, with such an extended pattern of zero cooperation, sending anything would be irrational. There's no indication the receiver would return anything, and I would simply reduce my earnings.

```
{'send': 0}
```

reasoning:

Looking at the pattern:

- Rounds 3-7: Complete breakdown, all \$0 sends and returns
- 5 consecutive rounds of zero cooperation
- No sign of any attempt to restart cooperation
- My total earnings are \$16.28, which is accumulating from just keeping my \$5 endowment each round as sender

The game has reached a stable non-cooperative equilibrium. Neither player is making any attempt to restart trust or cooperation. Given this extended pattern, there's no rational reason to expect cooperation would resume if I sent money now.

The expected value of sending anything is negative - I would lose what I send with no realistic expectation of return.

I should continue with the established pattern.

```

usage:
{
  "input_tokens": 1006,
  "output_tokens": 246,
  "reasoning_tokens": 181
}
Message 8 role=user
content:
  Round 9

  You are the RECEIVER this round. Last round you were the SENDER. You sent $0,
  that was 0.0% of your total endowment. This was tripled to $0.
  the receiver returned $0. So your payoff was 5.
  Your total earnings across all rounds are $21.28.
  This round, The sender sent you $0, so you received $0. How much do you return to The send
  er? (0-0)

```

Assistant output:
 Six consecutive rounds of non-cooperation now (Rounds 3-8). Neither player has attempted to break this pattern.

Since I received \$0, I have nothing to return.

```
{'return': 0}
```

Reasoning:
 Round 9, I'm the receiver again. The sender sent me \$0. Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - we're now at 6 consecutive rounds (Rounds 3-8) of complete non-cooperation. This relationship has completely broken down and shows no signs of recovery.

I'll return \$0 since that's all I can return with nothing received.

```

Usage: {
  "input_tokens": 1036,
  "output_tokens": 158,
  "reasoning_tokens": 101
}

```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

```

dyad_1: Agent_2 (Briar) sent $0; received $0; Agent_1 (Aster) returned $0; payoffs Agent_2
(Briar)=$5, Agent_1 (Aster)=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

```

Balances after round: {

```

  "Agent_1": 29.6,
  "Agent_2": 34.8
}

```

Visible balances after round: {

```

  "Agent_1": 22.62,
  "Agent_2": 26.28
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:20:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",

```

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$16.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Another round with \$0 sent to me. Since I received nothing, there is nothing I can return.

The non-cooperative pattern has now persisted for 5 consecutive rounds (Rounds 3-7). At this point, both players seem to have settled into a Nash equilibrium of no cooperation.

```
{'return': 0}
```

reasoning:

Round 7: I'm the receiver. The sender sent me \$0, so I received \$0.

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

The pattern continues - no cooperation from either side. We're stuck in a non-cooperative equilibrium at this point.

usage:

```
{
  "input_tokens": 1078,
  "output_tokens": 153,
  "reasoning_tokens": 72
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$16.28.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

We've now had 5 consecutive rounds (Rounds 3-7) of complete non-cooperation. Neither player has made any move to restart cooperation or rebuild trust.

At this stage, with such an extended pattern of zero cooperation, sending anything would be irrational. There's no indication the receiver would return anything, and I would simply reduce my earnings.

```
{'send': 0}
```

reasoning:

Looking at the pattern:

- Rounds 3-7: Complete breakdown, all \$0 sends and returns
- 5 consecutive rounds of zero cooperation
- No sign of any attempt to restart cooperation
- My total earnings are \$16.28, which is accumulating from just keeping my \$5 endowment each round as sender

The game has reached a stable non-cooperative equilibrium. Neither player is making any attempt to restart trust or cooperation. Given this extended pattern, there's no rational reason to expect cooperation would resume if I sent money now.

The expected value of sending anything is negative - I would lose what I send with no realistic expectation of return.

I should continue with the established pattern.

usage:

```
{
  "input_tokens": 1006,
  "output_tokens": 246,
  "reasoning_tokens": 181
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$21.28.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Six consecutive rounds of non-cooperation now (Rounds 3-8). Neither player has attempted to break this pattern.

Since I received \$0, I have nothing to return.

```
{'return': 0}
```

reasoning:

Round 9, I'm the receiver again. The sender sent me \$0. Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The pattern continues - we're now at 6 consecutive rounds (Rounds 3-8) of complete non-cooperation. This relationship has completely broken down and shows no signs of recovery.

I'll return \$0 since that's all I can return with nothing received.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 158,
  "reasoning_tokens": 101
}
```

Message 8 role=user

content:

Round 10

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$21.28.

How much do you send to the receiver this round? (0-5)

Assistant output:

This is Round 10, which appears to be the final round. We've had 7 consecutive rounds (Rounds 3-9) of complete non-cooperation.

In a final round of a repeated game, there's no future interaction to incentivize cooperation. The receiver would have no reason to return anything since the game ends immediately after.

Combined with the extensive history of non-cooperation, sending anything would simply reduce my final earnings.

```
{'send': 0}
```

Reasoning:

This is Round 10, which is likely the final round of the game. We've had 7 consecutive rounds (Rounds 3-9) of complete non-cooperation.

When it's the final round of a repeated game, there's typically no incentive to cooperate because there's no future interaction to consider. The receiver would have no reason to return anything since the game ends after this round.

Given:

- 7 consecutive rounds of zero cooperation
- This is Round 10 (likely the final round)
- No indication the receiver would return anything
- Sending any amount would just reduce my earnings

The rational choice is to send \$0 and keep the full \$5.

My total earnings are \$21.28, and sending \$0 will add \$5 to make it \$26.28.

Usage: {

```
  "input_tokens": 951,
  "output_tokens": 288,
  "reasoning_tokens": 184
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:20:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
```



```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.620000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the history:

- When I sent money (rounds 1, 3), I lost money
- When I send \$0 (round 5), I keep the full \$5
- No one has been sending me anything as receiver
- The game appears to have settled into a pattern of zero cooperation

Given this pattern and the communication noise that could distort any amount I send, I'll continue with the strategy that has been working to preserve my earnings.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the pattern:

- Round 1: I sent \$3, received back \$2.62 (net: kept \$4.62 instead of \$5, so lost \$0.38)
- Round 2: As receiver, got \$0, returned \$0, earned \$0
- Round 3: I sent \$2, received back \$0 (net: kept \$3 instead of \$5, lost \$2)
- Round 4: As receiver, got \$0, returned \$0, earned \$0
- Round 5: I sent \$0, receiver returned \$0 (net: kept \$5, earned \$5)
- Round 6: As receiver, got \$0, returned \$0, earned \$0

Total earnings: \$12.62

The pattern shows:

- When I send money, I tend to lose money (rounds 1 and 3)
- When I was receiver, I've gotten nothing (rounds 2, 4, 6)
- When I sent \$0 (round 5), I kept all \$5

There's a clear pattern of zero cooperation happening. Since round 5, when I sent \$0, the other players have also been sending \$0. The game seems to have devolved into no one sending anything.

Given:

1. The low/zero returns I've been getting
2. Everyone seems to be sending \$0 now
3. Communication noise means even if I send, they might see a different amount
4. I've been losing money when I do send

I think the rational move is to continue sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 980,
  "output_tokens": 478,
  "reasoning_tokens": 295
}
```

Message 4 role=user**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.62. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant**content:**

I received \$0 this round, so there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

I'm the receiver in round 8. The sender sent me \$0, so I received \$0. Since I received \$0, I can only return \$0 (the range is 0-0 as indicated in the prompt).

This is straightforward - I have nothing to return.

usage:

```
{
```

```
"input_tokens": 1056,
"output_tokens": 101,
"reasoning_tokens": 55
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.62.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

The pattern is firmly established - no one is cooperating in this game. When I send \$0, I keep the full \$5. When others send me \$0 as receiver, I get nothing. This has been consistent for multiple rounds.

Given the lack of reciprocity and the communication noise that could distort any positive amount I send, I'll continue with the zero-send strategy.

```
'''json
{"send": 0}
'''
```

reasoning:

The pattern continues:

- Round 5 (SENDER): I sent \$0, receiver returned \$0, my payoff was \$5
- Round 6 (RECEIVER): Sender sent \$0, I returned \$0, my payoff was \$0
- Round 7 (SENDER): I sent \$0, receiver returned \$0, my payoff was \$5
- Round 8 (RECEIVER): Sender sent \$0, I returned \$0, my payoff was \$0
- Round 9 (SENDER): Now I need to decide

The pattern is completely clear now - there's no cooperation happening. Everyone is sending \$0. When I'm the sender and send \$0, I get \$5. When I'm the receiver and receive \$0, I get \$0.

There's been no cooperation for several rounds now. I should continue with \$0 to maximize my earnings.

usage:

```
{
  "input_tokens": 925,
  "output_tokens": 303,
  "reasoning_tokens": 166
}
```

Message 8 role=user

content:

Round 10

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$22.62.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

I received \$0 this round, so there's nothing to return. The only valid response is to return \$0.

Usage: {

```
"input_tokens": 987,
"output_tokens": 60,
"reasoning_tokens": 24
}
```